

15. Line a has slope $\frac{3}{2}$ and line b has slope 4. Line ℓ bisects an angle formed by a and b and has positive slope. That slope can be written as $\frac{k+w\sqrt{p}}{q}$ where k , w , and q are relatively prime integers and $q > 0$. Determine the sum $(k+w+p+q)$.
16. Let T be the tetrahedron bounded by the points $(0,0,0)$, $(0,0,8)$, $(0,6,0)$, and $(10,0,0)$. The unit vector $\vec{V}$ normal to the face of the tetrahedron that contains the points $(0,0,8)$, $(0,6,0)$, and $(10,0,0)$ can be written as $\vec{V} = \left\langle \frac{x}{\sqrt{w}}, \frac{y}{\sqrt{w}}, \frac{z}{\sqrt{w}} \right\rangle$ with $x > 0$. Determine the sum $(x+y+z+w)$.
17. Let $p(x)$ and $q(x)$ be quartic polynomial functions with integral coefficients such that $p(1)=q(3)$, $p(2)=q(4)$, $p(3)=q(5)$, $p(4)=q(6)$, and $p(5)=q(7)+k$, where k is a positive multiple of 23. Determine the smallest value of k that satisfies these conditions.
18. In $\triangle ABC$ with right angle at C , $\frac{AC}{BA} = \frac{56}{65}$. $\tan \angle BAC = k$. Determine the value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
19. $\lim_{x \rightarrow \infty} \frac{4x + e^{(-x)}}{5x} = k$. Determine the value of k . Express your answer as an integer or a common or improper fraction.
20. A plane flying at an average speed of 420 mph at an angle of 40° west of due north encounters a wind pushing 20 mph at an angle of 20° west of due south. The plane will then be traveling at a new angle west of due north. Determine the degree measure of that angle. Express your answer as a decimal rounded to the nearest hundredth.